

4.204 Word embeddings

May 14, 2023

1 Word embeddings

1.1 An alternative approach

Can we define words by the company they keep?

“If A and B have almost the identical environments we can say that they are synonyms” (Zelig Harris, 1954)

1.2 Vector representations

One-hot encodings are long and sparse

Alternative: **dense vectors**

short (length 50-1000) + dense (most elements are non-zero) ### Benefits 1. Easier to use in ML
2. Offer better generalization capabilities

1.3 1. Train your own embeddings

Instead of counting terms, we train a classifier on a prediction task: ‘does A occur near B’?

The learned classifier weights become our embeddings

We can create our own word embeddings using gensim (an open-source library for unsupervised topic modeling and NLP)

and train it on the Brown corpus

```
[1]: # set up libraries & data
import gensim
from nltk.corpus import brown
import nltk
nltk.download('brown')
```

```
[nltk_data] Downloading package brown to /home/jovyan/nltk_data...
```

```
[nltk_data] Package brown is already up-to-date!
```

```
[1]: True
```

```
[2]: # train the model
model = gensim.models.Word2Vec(brown.sents())
```

```
[3]: # save a copy, for later re-use
model.save('brown.embedding')
```

```
[4]: # we can load models on demand
brown_model = gensim.models.Word2Vec.load('brown.embedding')
```

```
[5]: # how many words?
len(brown_model.wv.index_to_key)
```

```
[5]: 15173
```

```
[6]: # how many dimensions?
brown_model.wv['university']
```

```
[6]: array([ 0.11300982,  0.25615984,  0.20042767,  0.11220696, -0.06386014,
        -0.33227015,  0.20377511,  0.34550646, -0.30683893, -0.28182924,
         0.16876002, -0.2285828 ,  0.18933383,  0.16810797,  0.2431832 ,
        -0.16309226,  0.27026987, -0.14448091, -0.5239542 , -0.5166484 ,
         0.28470555, -0.11415465,  0.4844603 ,  0.07581545, -0.05946255,
        -0.13370386, -0.23097089,  0.01637265, -0.21084565,  0.23458743,
         0.2184081 , -0.06719489,  0.2808592 , -0.37119165, -0.17847078,
         0.05788324, -0.18882082, -0.05160563, -0.34307528, -0.0598388 ,
         0.01209599, -0.26870117,  0.1792478 ,  0.11543412,  0.20347296,
        -0.00693408, -0.01965668, -0.0248253 ,  0.10060211,  0.3107144 ,
         0.0160599 , -0.2868643 , -0.25124168, -0.18997039, -0.10733682,
        -0.22973788,  0.17746982,  0.0518054 , -0.06219205, -0.07913408,
         0.02912294,  0.20059325, -0.03721685, -0.16855565, -0.18875173,
         0.45090353,  0.02202957,  0.2841802 , -0.26235983,  0.3689759 ,
         0.2050297 ,  0.14639823,  0.21672332, -0.0026367 ,  0.33446604,
         0.08903393,  0.10674868,  0.04823826,  0.03011632, -0.13051347,
        -0.11879932,  0.01090001, -0.2168907 ,  0.07384715, -0.18467759,
        -0.01913904,  0.18961117,  0.01809401,  0.16601935,  0.07614642,
         0.42577514, -0.15034895, -0.11952778,  0.11187588,  0.32347074,
         0.12069657,  0.16820289, -0.36280492, -0.15385868,  0.05046168],
        dtype=float32)
```

```
[7]: # calculate similarity between terms
brown_model.wv.similarity('university','school')
```

```
[7]: 0.8153411
```

```
[8]: # find similar terms
brown_model.wv.most_similar('university', topn=5)
```

```
[8]: [('membership', 0.9541438221931458),
      ('profession', 0.9534937739372253),
      ('neighborhood', 0.9527076482772827),
```

```
('congregation', 0.951305627822876),  
( 'selection', 0.9480303525924683)]
```

```
[9]: brown_model.wv.most_similar('lemon', topn=5)
```

```
[9]: [('marble', 0.96579909324646),  
      ('pension', 0.9649028182029724),  
      ('frankfurters', 0.9640252590179443),  
      ('herd', 0.9626740217208862),  
      ('towel', 0.9624295234680176)]
```

```
[10]: brown_model.wv.most_similar('government', topn=5)
```

```
[10]: [('nation', 0.9279540181159973),  
      ('policy', 0.9250665903091431),  
      ('Christian', 0.923326849937439),  
      ('power', 0.9232913851737976),  
      ('education', 0.9203363656997681)]
```

```
[11]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair
```

```
brown_model.wv.most_similar('university', topn=5)
```

```
[11]: [('membership', 0.9541438221931458),  
      ('profession', 0.9534937739372253),  
      ('neighborhood', 0.9527076482772827),  
      ('congregation', 0.951305627822876),  
      ('selection', 0.9480303525924683)]
```

```
[12]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair
```

```
brown_model.wv.most_similar('college', topn=5)
```

```
[12]: [('university', 0.9234238266944885),  
      ('treatment', 0.9229620695114136),  
      ('selection', 0.9215680360794067),  
      ('persons', 0.9178371429443359),  
      ('generations', 0.9159908294677734)]
```

```
[13]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:
```

```
#university, college, school, factory, farm, pig, dog, cat, apple, lemon,
→table, chair
```

```
brown_model.wv.most_similar('school', topn=5)
```

```
[13]: [('college', 0.8936882615089417),
      ('special', 0.8661983609199524),
      ('service', 0.8564610481262207),
      ('training', 0.8519640564918518),
      ('private', 0.8354824781417847)]
```

```
[14]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      →following:
```

```
#university, college, school, factory, farm, pig, dog, cat, apple, lemon,
→table, chair
```

```
brown_model.wv.most_similar('factory', topn=5)
```

```
[14]: [('numerical', 0.9732374548912048),
      ('spectacular', 0.9698288440704346),
      ('gin', 0.9671325087547302),
      ('vigorous', 0.9660696387290955),
      ('establishments', 0.964354395866394)]
```

```
[15]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      →following:
```

```
#university, college, school, factory, farm, pig, dog, cat, apple, lemon,
→table, chair
```

```
brown_model.wv.most_similar('farm', topn=5)
```

```
[15]: [('aid', 0.9576115608215332),
      ('market', 0.9493377804756165),
      ('goods', 0.9455764889717102),
      ('standards', 0.9374123811721802),
      ('labor', 0.93666011095047)]
```

```
[16]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      →following:
```

```
#university, college, school, factory, farm, pig, dog, cat, apple, lemon,
→table, chair
```

```
brown_model.wv.most_similar('pig', topn=5)
```

```
[16]: [('pile', 0.9400528073310852),
      ('Vienna', 0.931883692741394),
      ('commissions', 0.9300448298454285),
```

```
('insert', 0.9277447462081909),  
('hay', 0.9275467991828918)]
```

```
[17]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
brown_model.wv.most_similar('dog', topn=5)
```

```
[17]: [('sign', 0.9510865211486816),  
      ('shelter', 0.9494808912277222),  
      ('reader', 0.9492417573928833),  
      ('cry', 0.9419220685958862),  
      ('attack', 0.9376896023750305)]
```

```
[18]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
brown_model.wv.most_similar('cat', topn=5)
```

```
[18]: [('pork', 0.9392828345298767),  
      ('sandals', 0.9391896724700928),  
      ('soloist', 0.9376404881477356),  
      ('meat', 0.9371135830879211),  
      ('lightning', 0.9354587197303772)]
```

```
[19]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
brown_model.wv.most_similar('apple', topn=5)
```

```
[19]: [('securely', 0.9413011074066162),  
      ('disguised', 0.9404338598251343),  
      ('replace', 0.9359477162361145),  
      ('insert', 0.9356834888458252),  
      ('thoroughly', 0.9343360662460327)]
```

```
[20]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair
```

```
brown_model.wv.most_similar('lemon', topn=5)
```

```
[20]: [('marble', 0.96579909324646),  
      ('pension', 0.9649028182029724),  
      ('frankfurters', 0.9640252590179443),  
      ('herd', 0.9626740217208862),  
      ('towel', 0.9624295234680176)]
```

```
[21]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
brown_model.wv.most_similar('table', topn=5)
```

```
[21]: [('hill', 0.9592476487159729),  
      ('window', 0.9565377831459045),  
      ('desk', 0.955863356590271),  
      ('edge', 0.9552389979362488),  
      ('wall', 0.9538423418998718)]
```

```
[22]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
brown_model.wv.most_similar('chair', topn=5)
```

```
[22]: [('nose', 0.9578806757926941),  
      ('hat', 0.9502779841423035),  
      ('shoulder', 0.9487252831459045),  
      ('lips', 0.9483065009117126),  
      ('shut', 0.9466347694396973)]
```

1.4 2. Use pre-trained embeddings

We can load pre-built embeddings, e.g. a sample from a model trained on 100 billion words from the Google News Dataset

```
[23]: import nltk  
      nltk.download('word2vec_sample')
```

```
[nltk_data] Downloading package word2vec_sample to  
[nltk_data] /home/jovyan/nltk_data...  
[nltk_data] Package word2vec_sample is already up-to-date!
```

[23]: True

```
[24]: # load a pre-build model
from nltk.data import find
word2vec_sample = str(find('models/word2vec_sample/pruned.word2vec.txt'))
news_model = gensim.models.KeyedVectors.load_word2vec_format(word2vec_sample,
↳ binary=False)
```

```
[25]: # how many terms?
len(news_model.key_to_index)
```

[25]: 43981

```
[26]: # how many dimensions?
len(news_model['university'])
```

[26]: 300

```
[27]: # are they any better?
news_model.most_similar(positive=['university'], topn = 5)
```

```
[27]: [('universities', 0.7003918290138245),
      ('faculty', 0.6780906915664673),
      ('undergraduate', 0.6587096452713013),
      ('campus', 0.6434987783432007),
      ('college', 0.638526976108551)]
```

```
[28]: news_model.most_similar(positive=['lemon'], topn = 5)
```

```
[28]: [('lemons', 0.646256148815155),
      ('apricot', 0.6199417114257812),
      ('avocado', 0.5922889113426208),
      ('fennel', 0.5873182415962219),
      ('coriander', 0.5828487277030945)]
```

```
[29]: news_model.most_similar(positive=['government'], topn = 5)
```

```
[29]: [('Government', 0.7132059335708618),
      ('governments', 0.6521531939506531),
      ('administration', 0.5462369322776794),
      ('legislature', 0.5307289361953735),
      ('parliament', 0.5268454551696777)]
```

```
[30]: #1. Use resources a) and b) to find the 5 most similar words for each of the
↳ following:
#university, college, school, factory, farm, pig, dog, cat, apple, lemon,
↳ table, chair
```

```
news_model.most_similar(positive=['university'], topn = 5)
```

```
[30]: [('universities', 0.7003918290138245),  
      ('faculty', 0.6780906915664673),  
      ('undergraduate', 0.6587096452713013),  
      ('campus', 0.6434987783432007),  
      ('college', 0.638526976108551)]
```

```
[31]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
news_model.most_similar(positive=['college'], topn = 5)
```

```
[31]: [('colleges', 0.6560819149017334),  
      ('university', 0.6385270357131958),  
      ('school', 0.6081898808479309),  
      ('collegiate', 0.6081600189208984),  
      ('undergraduate', 0.5866836905479431)]
```

```
[32]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
news_model.most_similar(positive=['school'], topn = 5)
```

```
[32]: [('elementary', 0.7868632078170776),  
      ('schools', 0.7411909699440002),  
      ('kindergarten', 0.6529811024665833),  
      ('School', 0.6477998495101929),  
      ('teacher', 0.63824063539505)]
```

```
[33]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair  
  
news_model.most_similar(positive=['factory'], topn = 5)
```

```
[33]: [('factories', 0.7827196717262268),  
      ('plant', 0.6708794236183167),  
      ('manufacturing', 0.6221225261688232),  
      ('Factory', 0.6064563393592834),  
      ('mill', 0.5385774970054626)]
```



```
[34]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      ↪following:
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,
      ↪table, chair
```

```
news_model.most_similar(positive=['farm'], topn = 5)
```

```
[34]: [('farms', 0.7599003314971924),
      ('farming', 0.7305765748023987),
      ('farmer', 0.710919201374054),
      ('Farm', 0.6409205198287964),
      ('agricultural', 0.6379074454307556)]
```

```
[35]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      ↪following:
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,
      ↪table, chair
```

```
news_model.most_similar(positive=['pig'], topn = 5)
```

```
[35]: [('pigs', 0.8023446798324585),
      ('swine', 0.7036389708518982),
      ('cow', 0.6542096734046936),
      ('goat', 0.6460473537445068),
      ('hog', 0.6289637088775635)]
```

```
[36]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      ↪following:
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,
      ↪table, chair
```

```
news_model.most_similar(positive=['dog'], topn = 5)
```

```
[36]: [('dogs', 0.868048906326294),
      ('puppy', 0.8106428384780884),
      ('cat', 0.760945737361908),
      ('pup', 0.7406911253929138),
      ('pet', 0.7164785265922546)]
```

```
[37]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      ↪following:
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,
      ↪table, chair
```

```
news_model.most_similar(positive=['cat'], topn = 5)
```

```
[37]: [('cats', 0.8099379539489746),  
      ('dog', 0.760945737361908),  
      ('kitten', 0.7464984059333801),  
      ('feline', 0.7326232194900513),  
      ('puppy', 0.7075453400611877)]
```

```
[38]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair
```

```
news_model.most_similar(positive=['apple'], topn = 5)
```

```
[38]: [('apples', 0.7203598022460938),  
      ('pear', 0.6450695991516113),  
      ('fruit', 0.6410147547721863),  
      ('berry', 0.6302295327186584),  
      ('pears', 0.6133959293365479)]
```

```
[39]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair
```

```
news_model.most_similar(positive=['lemon'], topn = 5)
```

```
[39]: [('lemons', 0.646256148815155),  
      ('apricot', 0.6199417114257812),  
      ('avocado', 0.5922889113426208),  
      ('fennel', 0.5873182415962219),  
      ('coriander', 0.5828487277030945)]
```

```
[40]: #1. Use resources a) and b) to find the 5 most similar words for each of the  
      ↪ following:  
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,  
      ↪ table, chair
```

```
news_model.most_similar(positive=['table'], topn = 5)
```

```
[40]: [('tables', 0.6950636506080627),  
      ('tray', 0.4653455317020416),  
      ('sideboard', 0.4434099793434143),  
      ('Tables', 0.4419552981853485),  
      ('napkin', 0.42967039346694946)]
```

```
[41]: #1. Use resources a) and b) to find the 5 most similar words for each of the
      ↪ following:
      #university, college, school, factory, farm, pig, dog, cat, apple, lemon,
      ↪ table, chair
```

```
news_model.most_similar(positive=['chair'], topn = 5)
```

```
[41]: [('chairs', 0.7871320247650146),
      ('chairman', 0.6161440014839172),
      ('chairing', 0.5431764125823975),
      ('Chairs', 0.5395590662956238),
      ('member', 0.4992404282093048)]
```

1.5 3. Perform vector algebra

We can use embeddings to perform verbal reasoning, e.g. A is to B as C is to...

e.g. 'man is to king as woman is to...'

$\text{vec}(\text{"king"}) - \text{vec}(\text{"man"}) + \text{vec}(\text{"woman"}) \approx \text{vec}(\text{"queen"})$

```
[42]: news_model.most_similar(positive=['woman', 'king'], negative=['man'], topn = 5)
```

```
[42]: [('queen', 0.7118193507194519),
      ('monarch', 0.6189674139022827),
      ('princess', 0.5902430415153503),
      ('prince', 0.5377321243286133),
      ('kings', 0.5236843228340149)]
```

```
[43]: news_model.most_similar(positive=['king', 'woman'], negative=['man'], topn = 5)
```

```
[43]: [('queen', 0.7118193507194519),
      ('monarch', 0.6189674139022827),
      ('princess', 0.5902430415153503),
      ('prince', 0.5377321243286133),
      ('kings', 0.5236843228340149)]
```

```
[44]: # encyclopaedic knowledge
      news_model.most_similar(positive=['Paris', 'Germany'], negative=['Berlin'], topn=
      ↪ 5)
```

```
[44]: [('France', 0.7884091734886169),
      ('Belgium', 0.6197876930236816),
      ('Spain', 0.5664774179458618),
      ('Italy', 0.5654898881912231),
      ('Switzerland', 0.560969352722168)]
```

```
[45]: # syntactic patterns (verbs)
news_model.most_similar(positive=['has', 'be'], negative=['have'], topn = 5)
```

```
[45]: [('is', 0.6774996519088745),
      ('was', 0.5710028409957886),
      ('remains', 0.47552669048309326),
      ('been', 0.4538103938102722),
      ('being', 0.4456518888473511)]
```

```
[46]: # syntactic patterns (adjectives)
news_model.most_similar(positive=['longest', 'short'], negative=['long'], topn = 5)
```

```
[46]: [('shortest', 0.5145131945610046),
      ('steepest', 0.42448338866233826),
      ('first', 0.40251171588897705),
      ('flattest', 0.40171927213668823),
      ('consecutive', 0.39518705010414124)]
```

```
[47]: # more encyclopaedic knowledge
news_model.most_similar(positive=['blue', 'tulip'], negative=['sky'], topn = 5)
```

```
[47]: [('purple', 0.5252774953842163),
      ('tulips', 0.4938238859176636),
      ('brown', 0.4907749593257904),
      ('pink', 0.4860530197620392),
      ('maroon', 0.48056456446647644)]
```

```
[48]: # syntactic knowledge (pronouns)
news_model.most_similar(positive=['him', 'she'], negative=['he'], topn = 5)
```

```
[48]: [('her', 0.804938554763794),
      ('herself', 0.6881042718887329),
      ('me', 0.5886672139167786),
      ('She', 0.5803765058517456),
      ('woman', 0.5470799207687378)]
```

```
[49]: # lexical know
news_model.most_similar(positive=['light', 'long'], negative=['dark'], topn = 5)
```

```
[49]: [('short', 0.4077243208885193),
      ('longer', 0.3670078217983246),
      ('lengthy', 0.36229580640792847),
      ('Long', 0.36003556847572327),
      ('continuous', 0.34982720017433167)]
```

```
[50]: # Find the odd one out
news_model.doesnt_match('breakfast cereal dinner lunch'.split())
```

```
[50]: 'cereal'
```

1.5.1 Quiz questions + homework

```
[51]: news_model.most_similar(positive=['be', 'has'], negative=['is'], topn = 5)
```

```
[51]: [('have', 0.7298133969306946),
      ('Had', 0.5585500597953796),
      ('subsequently', 0.54400634765625),
      ('had', 0.5378279685974121),
      ('recently', 0.5339969992637634)]
```

```
[52]: brown_model.wv.most_similar('university', topn=5)
```

```
[52]: [('membership', 0.9541438221931458),
      ('profession', 0.9534937739372253),
      ('neighborhood', 0.9527076482772827),
      ('congregation', 0.951305627822876),
      ('selection', 0.9480303525924683)]
```

```
[53]: news_model.most_similar('university', topn=5)
```

```
[53]: [('universities', 0.7003918290138245),
      ('faculty', 0.6780906915664673),
      ('undergraduate', 0.6587096452713013),
      ('campus', 0.6434987783432007),
      ('college', 0.638526976108551)]
```

```
[54]: brown_model.wv.most_similar('college', topn=5)
```

```
[54]: [('university', 0.9234238266944885),
      ('treatment', 0.9229620695114136),
      ('selection', 0.9215680360794067),
      ('persons', 0.9178371429443359),
      ('generations', 0.9159908294677734)]
```

```
[55]: news_model.most_similar('college', topn=5)
```

```
[55]: [('colleges', 0.6560819149017334),
      ('university', 0.6385270357131958),
      ('school', 0.6081898808479309),
      ('collegiate', 0.6081600189208984),
      ('undergraduate', 0.5866836905479431)]
```

```
[56]: #2. Use resources b) and c) to rank the degrees of similarity between the word
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle
```

```
brown_model.wv.similarity('university','college')
```

```
[56]: 0.92342395
```

```
[57]: #2. Use resources b) and c) to rank the degrees of similarity between the word
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle
```

```
brown_model.wv.similarity('university','school')
```

```
[57]: 0.8153411
```

```
[58]: #2. Use resources b) and c) to rank the degrees of similarity between the word
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle
```

```
brown_model.wv.similarity('university','factory')
```

```
[58]: 0.83194214
```

```
[59]: #2. Use resources b) and c) to rank the degrees of similarity between the word
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle
```

```
brown_model.wv.similarity('university','supermarket')
```

```

      ↪ -----

      KeyError                                Traceback (most recent call
      ↪ last)

      <ipython-input-59-c5c71ebfc570> in <module>
          2 #college, school, factory, supermarket, turtle
          3
      ----> 4 brown_model.wv.similarity('university','supermarket')

      /opt/conda/lib/python3.7/site-packages/gensim/models/keyedvectors.py in
      ↪ similarity(self, w1, w2)
          1153
          1154 """
```

```

-> 1155         return dot(matutils.unitvec(self[w1]), matutils.
↪unitvec(self[w2]))
    1156
    1157     def n_similarity(self, ws1, ws2):

/opt/conda/lib/python3.7/site-packages/gensim/models/keyedvectors.py in
↪__getitem__(self, key_or_keys)
    377         """
    378         if isinstance(key_or_keys, KEY_TYPES):
--> 379             return self.get_vector(key_or_keys)
    380
    381         return vstack([self.get_vector(key) for key in key_or_keys])

/opt/conda/lib/python3.7/site-packages/gensim/models/keyedvectors.py in
↪get_vector(self, key, norm)
    420
    421         """
--> 422         index = self.get_index(key)
    423         if norm:
    424             self.fill_norms()

/opt/conda/lib/python3.7/site-packages/gensim/models/keyedvectors.py in
↪get_index(self, key, default)
    394         return default
    395     else:
--> 396         raise KeyError(f"Key '{key}' not present")
    397
    398     def get_vector(self, key, norm=False):

KeyError: "Key 'supermarket' not present"

```

```

[60]: #2. Use resources b) and c) to rank the degrees of similarity between the word
↪ 'university' and each of the following:
#college, school, factory, supermarket, turtle

brown_model.wv.similarity('university', 'turtle')

```

[60]: 0.8078926

```

[61]: #2. Use resources b) and c) to rank the degrees of similarity between the word
↪ 'university' and each of the following:
#college, school, factory, supermarket, turtle

```

```
news_model.similarity('university','college')
```

[61]: 0.638527

```
[62]: #2. Use resources b) and c) to rank the degrees of similarity between the word_
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle

      news_model.similarity('university', 'school')
```

[62]: 0.5080746

```
[63]: #2. Use resources b) and c) to rank the degrees of similarity between the word_
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle

      news_model.similarity('university', 'factory')
```

[63]: 0.13740103

```
[64]: #2. Use resources b) and c) to rank the degrees of similarity between the word_
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle

      news_model.similarity('university', 'supermarket')
```

[64]: 0.13598306

```
[65]: #2. Use resources b) and c) to rank the degrees of similarity between the word_
      ↪ 'university' and each of the following:
      #college, school, factory, supermarket, turtle

      news_model.similarity('university', 'turtle')
```

[65]: 0.054503173

[]: